

A quartic product-prior counterexample

to the general small-ball ratio conjecture

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Abstract

We give a proposed counterexample to the unresolved general statement in AIM Problem 506, equivalently Conjecture 4.12 of Ayanbayev, Klebanov, Lie and Sullivan. On ℓ^2 we construct a countable product of scaled copies of the density $\rho(z) \propto e^{-z^4}$ and an explicit center h with formal energy $Q(h) = 1$, but

$$\liminf_{r \downarrow 0} \frac{\mu(B_r(h))}{\mu(B_r(0))} = 0 \neq e^{-1}.$$

The density is positive, real analytic and log-concave, with finite Fisher information. A one-dimensional Gaussian-convolution inequality and a Laplace-transform averaging identity give a self-contained analytic disproof. We also give explicit bounds on a vanishing sequence of ball ratios, arbitrarily small-energy counterexamples, and a second example with polynomial coordinate scales. Exact-arithmetic certificates and separate numerical checks are supplementary, not prerequisites for the main theorem. No assertion that the full small-ball ratio limit exists is needed or made.

1 Target, construction and admissibility

AIM Problem 506 [2] asks, for scaled symmetric product priors on weighted ℓ^p spaces, whether

$$\lim_{r \downarrow 0} \frac{\mu(B_r(h))}{\mu(B_r(m))} = e^{-Q(h)}, \quad Q(h) = \sum_{k=1}^{\infty} \log \frac{\rho(0)}{\rho((h_k - m_k)/\gamma_k)}. \quad (1)$$

This is Conjecture 4.12 of [1], under its Assumption 4.1(A1)–(A3). The repository labels the entry *Partial*: several cases are proved, but the unrestricted finite-energy lower bound remains open. The example below addresses that remaining target, rather than reproving a known case.

Theorem 1.1 (Explicit counterexample). *Let $X = \ell^2(\mathbb{N})$, $m = 0$, and let the independent Z_k have density*

$$\rho(z) = Z_0^{-1} e^{-z^4}, \quad Z_0 = \int_{\mathbb{R}} e^{-z^4} dz = \frac{1}{2} \Gamma(1/4). \quad (2)$$

Partition $\mathbb{N}$ into consecutive blocks I_j of cardinality 8^j , $j \geq 1$. On I_j , set

$$\gamma_k = 4^{-j}, \quad h_k = 8^{-j}, \quad d_k := h_k/\gamma_k = 2^{-j}. \quad (3)$$

Write $Y = (\gamma_k Z_k)_{k \geq 1}$ and $\mu = \text{Law}(Y)$. Then μ is a probability measure with full support on ℓ^2 , $h \in \ell^2$, and

$$Q(h) = 1, \quad \|h\|_2^2 = \frac{1}{7}, \quad \liminf_{r \downarrow 0} \frac{\mu(B_r(h))}{\mu(B_r(0))} = 0. \quad (4)$$

Consequently (1) is false under its stated assumptions.

1.1 Verification of all hypotheses

The block endpoints may be taken as $M_0 = 0$ and $M_j = \sum_{i=1}^j 8^i = 8(8^j - 1)/7$, with $I_j = \{M_{j-1} + 1, \dots, M_j\}$. The parameters in the repository are $p = 2$ and $\alpha_k = 1$. The density in (2) is even, positive and strictly decreasing on $[0, \infty)$. It is real analytic and log-concave. Every polynomial moment is finite. Integration of $(ze^{-z^4})'$ over $\mathbb{R}$ gives $\mathbb{E}Z^4 = 1/4$, so Cauchy–Schwarz yields $\mathbb{E}Z^2 \leq 1/2$.

All relevant series are explicit:

$$\sum_k \gamma_k^2 = \sum_{j=1}^{\infty} 8^j 16^{-j} = 1, \quad (5)$$

$$\|h\|_2^2 = \sum_{j=1}^{\infty} 8^j 64^{-j} = \frac{1}{7}, \quad (6)$$

$$Q(h) = \sum_k d_k^4 = \sum_{j=1}^{\infty} 8^j 16^{-j} = 1, \quad (7)$$

$$\sum_k d_k^2 = \sum_{j=1}^{\infty} 2^j = \infty. \quad (8)$$

Tonelli's theorem and (5) imply $\mathbb{E} \sum_k \gamma_k^2 Z_k^2 \leq 1/2$, hence the random sequence is in ℓ^2 almost surely. It is an ℓ^2 -valued measurable random variable, being the almost-sure norm limit of its finite-coordinate truncations.

For completeness, every open ball centered at $x \in \ell^2$ has positive mass. Fix $r > 0$ and choose a coordinate cutoff M such that $\|x_{>M}\|_2 < r/4$ and $\mathbb{E} \|Y_{>M}\|_2^2 < r^2/32$. Markov's inequality gives $\mathbb{P}(\|Y_{>M}\|_2 < r/4) > 1/2$. The first M coordinates have a positive joint density, so $\mathbb{P}(\|Y_{\leq M} - x_{\leq M}\|_2 < r/2) > 0$. Independence and the triangle inequality show $\mu(B_r(x)) > 0$.

There is no lack-of-smoothness issue: $\rho' = -4z^3\rho$ and $\rho'' = (16z^6 - 12z^2)\rho$, so

$$\int_{\mathbb{R}} \frac{(\rho')^2}{\rho} dz = 16\mathbb{E}Z^6 < \infty, \quad \rho'' \in L^1(\mathbb{R}).$$

Thus the example even satisfies the optional regularity assumptions (A4) and (A5) of [1]. Its standardized displacement is not square summable, as (8) records.

2 A one-dimensional convolution inequality

For $a > 0$ and $d \in \mathbb{R}$, define

$$\ell(a, d) = \mathbb{E}e^{-a(Z-d)^2}, \quad R_a(d) = \frac{\ell(a, d)}{\ell(a, 0)}, \quad f_a(d) = \int_{\mathbb{R}} e^{-z^4 - a(z-d)^2} dz. \quad (9)$$

The normalization Z_0 cancels in $R_a(d) = f_a(d)/f_a(0)$.

Lemma 2.1 (Centered convolution and a quadratic penalty). *For every $a > 0$ and $d \in \mathbb{R}$,*

$$e^{-ad^2} \leq R_a(d) \leq 1. \quad (10)$$

Moreover, for $|d| \leq 1$,

$$R_1(d) \leq \exp(-d^2/144). \quad (11)$$

Proof. The convolution of two even nonincreasing functions is maximized at zero; here this can be seen without a rearrangement theorem. Write $e^{-a(z-d)^2}$ by its level sets. Each level set is an interval centered at d , and, for $r \geq 0$, the integral of the even decreasing density e^{-z^4} over $[d-r, d+r]$ is maximal at $d = 0$. For $d \geq 0$ its derivative is $e^{-(d+r)^4} - e^{-(d-r)^4} \leq 0$. Tonelli's theorem proves $f_a(d) \leq f_a(0)$.

Let $\eta_{a,0}$ be the probability measure proportional to $e^{-z^4 - az^2} dz$ and let $\eta_{a,d}$ be its tilt proportional to e^{2adz} . By symmetry,

$$R_a(d) = e^{-ad^2} \mathbb{E}_{\eta_{a,0}} \cosh(2adZ) \geq e^{-ad^2}. \quad (12)$$

All differentiations below are dominated on compact parameter sets by an integrable quartic exponential times a polynomial. Integration of the derivative in z gives

$$4\mathbb{E}_{\eta_{a,d}} Z^3 + 2a(\mathbb{E}_{\eta_{a,d}} Z - d) = 0,$$

so

$$(\log f_a)'(d) = 2a(\mathbb{E}_{\eta_{a,d}} Z - d) = -4\mathbb{E}_{\eta_{a,d}} Z^3. \quad (13)$$

Set $\kappa(a) = \mathbb{E}_{\eta_{a,0}} Z^4$. For $d \geq 0$,

$$\mathbb{E}_{\eta_{a,0}} [Z^3 e^{2adZ}] = \mathbb{E}_{\eta_{a,0}} [Z^3 \sinh(2adZ)] \geq 2ad \kappa(a).$$

The inequality follows pointwise from $t \sinh t \geq t^2$. Also,

$$\mathbb{E}_{\eta_{a,0}} e^{2adZ} = e^{ad^2} R_a(d) \leq e^{ad^2}.$$

Substituting into (13) and integrating from 0 to d yields, by evenness also for negative d ,

$$\log R_a(d) \leq -4\kappa(a)(1 - e^{-ad^2}). \quad (14)$$

It remains to bound $\kappa := \kappa(1)$ by an elementary positive constant. The denominator $\int_{\mathbb{R}} e^{-z^4 - z^2} dz$ is smaller than $\sqrt{\pi} < 2$. On $[-1, -1/2] \cup [1/2, 1]$, a set of total length 1, one has $z^4 \geq 1/16$ and $z^4 + z^2 \leq 2$. Hence the numerator is at least $e^{-2}/16 > 1/144$, since $e < 3$. Therefore

$$\kappa > \frac{1}{288}. \quad (15)$$

For $0 \leq u \leq 1$, Taylor's inequality gives $1 - e^{-u} \geq u - u^2/2 \geq u/2$. Applying this with $u = d^2$ in (14) proves (11). $\square$

Remark 2.2 (Why a quartic density can generate a quadratic cost). A second integration by parts gives $1 = 4\kappa + 2\mathbb{E}_{\eta_{1,0}} Z^2$. Expanding the analytic even function $\log R_1(d)$ at zero therefore gives

$$\log R_1(d) = -4\kappa d^2 + O(d^4).$$

The formal density cost is d^4 , but Gaussian convolution at the active coordinate scale produces a strictly positive quadratic displacement cost. This mismatch drives the construction.

3 Laplace transforms and the small-ball contradiction

For $x \in \ell^2$ and $\tau > 0$, define

$$L_x(\tau) = \mathbb{E} e^{-\tau \|Y - x\|_2^2}. \quad (16)$$

Since the random norm is finite almost surely, $0 < L_x(\tau) \leq 1$. Independence gives factorization over finite coordinate prefixes. Dominated convergence then gives, for the full product,

$$\frac{L_h(\tau)}{L_0(\tau)} = \prod_{j=1}^{\infty} R_{\tau 16^{-j}}(2^{-j})^{8^j}. \quad (17)$$

This identity is the limit of ratios of finite products whose denominators converge to the positive number $L_0(\tau)$; no vanishing denominator occurs in this limit.

Every factor in (17) lies in $(0, 1]$. At the selected scale $\tau_j = 16^j$, Lemma 2.1 applied to the j th block gives

$$0 < \frac{L_h(16^j)}{L_0(16^j)} \leq R_1(2^{-j})^{8^j} \leq \exp\left(-\frac{8^j 4^{-j}}{144}\right) = \exp(-2^j/144) \rightarrow 0. \quad (18)$$

The following lemma turns this statement into the required small-ball contradiction.

Lemma 3.1 (Laplace averaging detects a vanishing ball-ratio liminf). *Let μ be a probability measure on a normed space with $\mu(B_r(0)) > 0$ for every $r > 0$, and let h belong to that space. Define L_x by (16). If, along some sequence $\tau_j \rightarrow \infty$, $L_h(\tau_j)/L_0(\tau_j) \rightarrow 0$, then*

$$\liminf_{r \downarrow 0} \frac{\mu(B_r(h))}{\mu(B_r(0))} = 0.$$

Proof. Tonelli's theorem and the identity $e^{-\tau s^2} = \int_s^\infty 2\tau r e^{-\tau r^2} dr$ give

$$L_x(\tau) = \int_0^\infty 2\tau r e^{-\tau r^2} \mu(B_r(x)) dr. \quad (19)$$

Open versus closed balls do not affect this identity, since the possible equality $r = s$ is a single Lebesgue-null point for each fixed s . Define the probability measure

$$d\omega_\tau(r) = \frac{2\tau r e^{-\tau r^2} \mu(B_r(0))}{L_0(\tau)} dr \quad (r > 0), \quad g(r) = \frac{\mu(B_r(h))}{\mu(B_r(0))}.$$

Then

$$\frac{L_h(\tau)}{L_0(\tau)} = \int_0^\infty g(r) d\omega_\tau(r). \quad (20)$$

For any $\varepsilon > 0$,

$$\begin{aligned} \omega_\tau([\varepsilon, \infty)) &\leq \frac{e^{-\tau \varepsilon^2}}{L_0(\tau)} \\ &\leq \frac{e^{-3\tau \varepsilon^2/4}}{\mu(B_{\varepsilon/2}(0))} \rightarrow 0. \end{aligned} \quad (21)$$

The last inequality follows by restricting $L_0(\tau)$ to $B_{\varepsilon/2}(0)$. If $\liminf_{r \downarrow 0} g(r) > 0$, there exist $c, \varepsilon > 0$ such that $g(r) \geq c$ whenever $0 < r < \varepsilon$. Nonnegativity in (20) gives $L_h(\tau)/L_0(\tau) \geq c \omega_\tau((0, \varepsilon))$, hence $\liminf_{\tau \rightarrow \infty} L_h(\tau)/L_0(\tau) \geq c$, contradicting the assumed zero subsequence. Since $g \geq 0$, its liminf must be zero. $\square$

Completion of Theorem 1.1. Admissibility, full support and $Q(h) = 1$ were proved in Section 1. Apply Lemma 3.1 to (18). The resulting liminf is 0, whereas the conjecture prescribes the positive limit e^{-1} . $\square$

Remark 3.2 (Exact scope). The argument proves a complete disproof of the universal limit formula. It does not establish that the full ball-ratio limit for this example is zero, and does not need that stronger claim. No Tauberian converse is being asserted: the only inference is the positive-liminf contradiction in Lemma 3.1.

4 Quantitative and robust forms

4.1 Explicit shrinking bounds on counterexample radii

Proposition 4.1. *For every integer $j \geq 1$, some radius r_j satisfies*

$$0 < r_j < 3 \cdot 2^{-j}, \quad \frac{\mu(B_{r_j}(h))}{\mu(B_{r_j}(0))} < 2e^{-4^j/144}. \quad (22)$$

In particular $r_j \rightarrow 0$. No specific value of r_j is required or claimed.

Proof. We first show

$$L_0(16^J) \geq e^{-5 \cdot 8^J} \quad (J \geq 1). \quad (23)$$

The normalizer Z_0 is smaller than 3: on $[-1, 1]$ its integral is at most 2, and outside this interval $e^{-z^4} \leq e^{-|z|}$ gives a tail at most $2/e < 1$. For $a \geq 1$, restricting the defining integral to $|z| \leq a^{-1/2}$ gives

$$\ell(a, 0) \geq \frac{2}{27} a^{-1/2} \geq e^{-3} a^{-1/2},$$

using $e < 3$ and $e^3 > 27/2$. For a block $i \leq J$, $a = 16^{J-i}$ and hence $-\log \ell(a, 0) \leq 3 + 2(J - i)$, since $\log 4 < 2$. The total contribution of these blocks is bounded by

$$8^J \sum_{l=0}^{\infty} 8^{-l} (3 + 2l) = \frac{184}{49} 8^J.$$

For $i > J$, Jensen's inequality and $\mathbb{E}Z^2 \leq 1/2$ give $-\log \ell(16^{J-i}, 0) \leq 16^{J-i}/2$. Their total contribution is at most

$$\frac{1}{2} \sum_{i>J} 8^i 16^{J-i} = \frac{1}{2} 8^J.$$

The sum of the two constants is $417/98 < 5$, proving (23).

Now choose $\tau = 16^{2j}$ and $\varepsilon_j = 3 \cdot 2^{-j}$ in the first line of (21). Equation (23) yields

$$\omega_{16^{2j}}([\varepsilon_j, \infty)) \leq e^{-9 \cdot 64^j + 5 \cdot 64^j} = e^{-4 \cdot 64^j} < \frac{1}{2}.$$

Equation (18) bounds the $\omega_{16^{2j}}$ -average of g by $e^{-4^j/144}$. If $g(r) \geq 2e^{-4^j/144}$ for every $0 < r < \varepsilon_j$, that average would be strictly larger than $e^{-4^j/144}$, a contradiction. This proves (22). $\square$

4.2 Arbitrarily small formal energy

For every $\eta > 0$, replace h by ηh in the same measure. Then

$$Q(\eta h) = \eta^4, \quad \|\eta h\|_2^2 = \eta^2/7.$$

For all sufficiently large j , $\eta 2^{-j} \leq 1$, and the selected block estimate becomes

$$\frac{L_{\eta h}(16^j)}{L_0(16^j)} \leq e^{-\eta^2 2^j/144} \longrightarrow 0.$$

Lemma 3.1 again gives a zero ball-ratio liminf, although the conjectured answer is $e^{-\eta^4} > 0$. Thus imposing merely a small bound on the formal energy does not repair the conjecture.

4.3 A second example with polynomial scales

The block construction is not essential. With the same reference density and $X = \ell^2$, take

$$\gamma_k = k^{-1}, \quad h_k = k^{-4/3}, \quad d_k = k^{-1/3}. \quad (24)$$

Then $\sum \gamma_k^2 < \infty$, $h \in \ell^2$, and $Q(h) = \sum_k k^{-4/3} = \zeta(4/3) < \infty$. Support and measurability follow as before.

To obtain a uniform version of the convolution estimate, note that

$$\kappa'(a) = -\text{Cov}_{\eta_{a,0}}(Z^4, Z^2) \leq 0.$$

Indeed, for independent nonnegative U, U' with the law of Z^2 , twice the covariance of U^2 and U equals $\mathbb{E}[(U - U')^2(U + U')] \geq 0$. Thus $\kappa(a) \geq \kappa(1) > 1/288$ for $1/4 \leq a \leq 1$. Equation (14) gives, for these a and $|d| \leq 1$,

$$\log R_a(d) \leq -2\kappa(a)ad^2 \leq -d^2/576. \quad (25)$$

At $\tau = n^2$, the n coordinates $n \leq k < 2n$ satisfy $1/4 \leq \tau\gamma_k^2 \leq 1$. Keeping just these factors in the Laplace ratio yields

$$\frac{L_h(n^2)}{L_0(n^2)} \leq \exp\left(-\frac{1}{576} \sum_{k=n}^{2n-1} k^{-2/3}\right) \leq \exp\left(-\frac{2^{-2/3}}{576} n^{1/3}\right) \rightarrow 0.$$

This proves a second counterexample to the conjecture using the single smooth power-law scale sequence (24).

5 Relation to established results and evidence status

Theorem 4.10 of [1], printed p.11, proves the general upper limsup bound. Under its smoothness assumption (A5), its lower bound is stated for finite-energy centers in $m + \ell_\gamma^2$. The conjecture on printed p.12 removes that center restriction. Our example satisfies (A5), but (8) shows that its center lies outside ℓ_γ^2 ; hence it does not contradict the established theorem. It shows that smoothness, log-concavity and finite Fisher information alone do not justify removal of that restriction.

The application stated in [2, 1] concerns using small-ball ratios to identify an Onsager–Machlup functional for product priors in Bayesian inverse problems. The result here rules out identifying that functional with the formal quartic energy on *all* finite-energy centers under the stated general assumptions. It does not rule out restricted-domain formulas, the known Gaussian cases, or all forms of maximum a posteriori estimation.

The main-branch problem entry and the pinned revision were checked, together with public submission records [3]. The current audit inspected all indexed issues and pull requests, available comments, reviews and file changes, the main branch, and focused subject searches. No matching AIM 506 submission was found. The dated audit accompanies this package and states its coverage limits. A bounded literature search found no same-target resolution; this does not establish exhaustive priority.

The supplied argument is AI-generated. Codex and separate Codex AI reviewers checked the complete argument, primary-source match and verification programs during preparation. No human peer review or proof-assistant verification is claimed; the accompanying review record states the scope. Under the repository’s evidence rules [4], the appropriate proposed status is *Solution claimed*. This submission contains the proof package; no catalogue status change is proposed. The analytic proof, exact supporting certificate and floating-point experiments must be distinguished.

A Optional exact-arithmetic integral certificate

The main proof uses only the elementary bound $\kappa > 1/288$. The supplied standard-library program `verify_exact.py` additionally encloses the integrals

$$A_r = \int_{\mathbb{R}} z^r e^{-z^4 - z^2} dz \quad (r = 0, 4)$$

with rational arithmetic and certifies $\kappa = A_4/A_0 > 1/8$. This strengthens (11) to $R_1(d) \leq e^{-d^2/4}$ for $|d| \leq 1$, but is not needed for the disproof.

Here is the entire enclosure method. For $u \geq 0$, Taylor's remainder implies $S_{101}(u) \leq e^{-u} \leq S_{100}(u)$, where $S_N(u) = \sum_{m=0}^N (-u)^m/m!$. Integrating over $[-2, 2]$ gives the exact rational numbers

$$T_{r,N} = \sum_{m=0}^N \frac{(-1)^m}{m!} \sum_{k=0}^m \binom{m}{k} \frac{2^{r+2m+2k+2}}{r+2m+2k+1}.$$

For $P(x) = x^4 + x^2$ and $x \geq 2$, convexity gives $P(x) \geq 20 + 36(x-2)$, while $x^4 \leq 16e^{2(x-2)}$ follows from $\log(x/2) \leq (x-2)/2$. Thus the two-sided tails satisfy

$$\int_{|x|>2} e^{-P(x)} dx \leq \frac{e^{-20}}{18}, \quad \int_{|x|>2} x^4 e^{-P(x)} dx \leq \frac{16e^{-20}}{17}.$$

An upper bound on e^{-20} is obtained by taking the reciprocal of a finite positive Taylor lower bound for e^{20} . Consequently,

$$\begin{aligned} A_0 &\in [T_{0,101}, T_{0,100} + e^{-20}/18], \\ A_4 &\in [T_{4,101}, T_{4,100} + 16e^{-20}/17], \end{aligned}$$

with the exponential upper bounds substituted as rational numbers.

The exact certificate in `exact_results.json` proves

$$0.13302001 < \kappa < 0.13302003, \quad \kappa > 1/8.$$

All certificate decisions use integers and fractions; decimal conversions are for display only. The recorded run passes 31 supporting checks, including block arithmetic, geometric series, the quantitative radius constants and the integral enclosures. This does not constitute a formal verification of the measure-theoretic proof.

B Numerical cross-checks and reproduction

The separate script `verify_numerical.py` uses 60-digit `mpmath` quadrature and a direct convolution implementation with `scipy.integrate.quad`. The recorded run passes 80 numerical checks. Across 25 comparisons, the maximum absolute discrepancy of the logarithmic one-dimensional ratios is below the portable tolerance 3×10^{-12} . The current machine-readable report records the measured maximum.

For example, the following are logarithms of *single-block factors*, not direct measurements of small-ball probabilities:

j	$d = 2^{-j}$	$\log(R_1(2^{-j})^{8^j})$
1	1/2	-1.0742745865383273
2	1/4	-2.1334827031180223
4	1/16	-8.5145804142510394
6	1/64	-34.0534502189846013
8	1/256	-136.2125824831589036

The script also computes numerical prefixes of the full Laplace product. From (10), the omitted logarithmic tail beyond block K obeys the analytic bound

$$-\frac{\tau 8^{-K}}{7} \leq \sum_{i>K} 8^i \log R_{\tau 16^{-i}}(2^{-i}) \leq 0.$$

The quadrature of the retained prefix is *not* certified interval arithmetic, so those floating-point values remain corroborations, even though the omitted-coordinate bound is analytic. No finite parameter grid is used as a substitute for the proof.

From the extracted package directory, run:

```
python3 verify_exact.py --output exact_results.json
python3 -m pip install -r requirements-numerical.txt
python3 verify_numerical.py --output numerical_results.json
```

The first command requires only Python’s standard library. The second command installs optional numerical-check dependencies. To rebuild this note:

```
pdflatex -interaction=nonstopmode -halt-on-error aim506_counterexample.tex
pdflatex -interaction=nonstopmode -halt-on-error aim506_counterexample.tex
```

The package also includes a target audit, a resolution report, environment versions and file hashes. Neither a proof assistant nor an independent human review has verified this result.

References

- [1] B. Ayanbayev, I. Klebanov, H. C. Lie and T. J. Sullivan, *Γ -convergence of Onsager–Machlup functionals: II. Infinite product measures on Banach spaces*, Inverse Problems **38** (2022), 025006. doi:10.1088/1361-6420/ac3f82. See Assumption 4.1, Theorem 4.10 and Conjecture 4.12, printed pp.6, 11–12. arXiv:2108.04598, version 3, 29 November 2021.
- [2] MColbrook/AIM, *Problem 506: Small-ball ratios for general symmetric product priors*. Pinned repository entry; main-branch entry also inspected. Repository status: Partial; entry’s last review: 23 September 2026.
- [3] MColbrook/AIM, *Public pull-request list*, inspected during this session on 2 October 2026. <https://github.com/MColbrook/AIM/pulls?page=1>. This is a bounded public-status check, not an exhaustive literature registry.
- [4] MColbrook/AIM, *Contributing guidelines*, revision aa776a01d7d48a79f93251af11fde9454b0aea95. Evidence requirements and status definitions.